PHY101: Introduction to Physics-1

Monsoon, 2017

Tutorial-10

1. A 10 kg mass is attached to a spring, stretching it 0.7 m from its natural length. The mass is started in motion from the equilibrium position with an initial velocity of 1 m/sec in the upward direction. Find the subsequent motion, if the force due to air resistance is -90$\dot{x}$ N.

2. A particle of mass m moves along the x-axis and is acted upon by the restoring force, $F=-(n^2+k^2)x$ and the resistance force, $f=-2mk\dot{x}$, where n, k are positive constants. If the particle is released from rest at $x = a$, show that, in the subsequent motion,

$$x = \frac{a}{n}e^{\frac{k}{t}}(ncosnt + ksinnt).$$

Find how far the particle travels before it next comes to rest.

3. The motion of a damped oscillator driven by an applied force $F_0 \cos \omega t$ is given by $x_a(t) = X_0 \cos(\omega t + \varphi)$, where X_0 and φ have their usual meaning. Consider an oscillator that is released from rest at $t = 0$. Its motion must satisfy $x(0) = 0$, $v(0) = 0$, but after a very long time, we expect that $x(t) = x_a(t)$. To satisfy these conditions we can take as the solution

$$x(t) = x_a(t) + x_b(t),$$

where $x_b(t)$ is the solution to the equation of motion of the free damped oscillator.

(*a*) Show that if $x_a(t)$ satisfies the equation of motion for the driven damped oscillator, then so does $x(t) = x_a(t) + x_b(t)$, where $x_b(t)$ satisfies the equation of motion of the free damped oscillator.

(*b*)Choose the arbitrary constants in $x_b(t)$ so that $x(t)$ satisfies the initial conditions. (Note that A and φ here are arbitrary.)